VIII. Mini Project-Supermarket Sales Analytics

AIM:

To clean, process, analyze, and visualize supermarket sales datasets using Python, and create dashboard-based visualizations using Power BI and Tableau to derive meaningful insights into product performance, branch sales, customer behavior, payment methods, and sales trends.

DOMAIN:

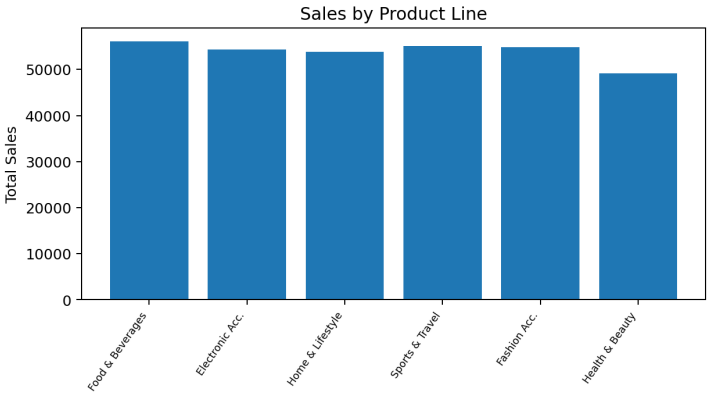
Retail Analytics / Sales Analytics

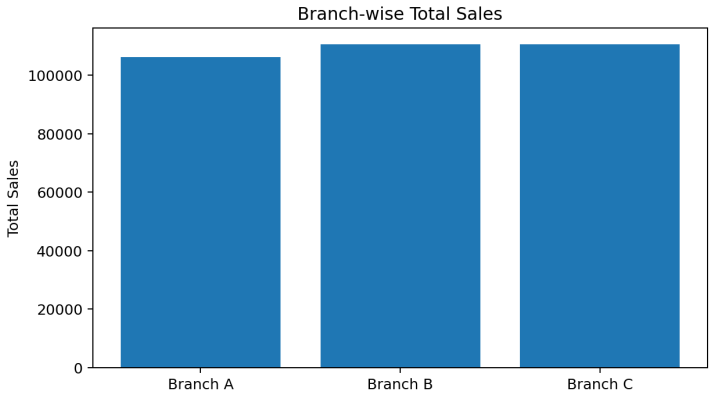
PROBLEM STATEMENT:

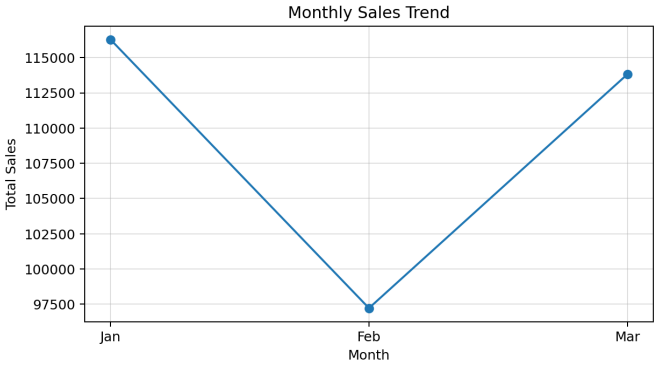
Supermarkets generate a large amount of transaction data containing information about branches, product lines, customers, quantity, total sales, payment methods, and ratings. Analyzing this data manually makes it difficult to identify sales patterns and business trends. Therefore, there is a need for a data-driven visualization system that processes supermarket data and presents meaningful insights through statistical analysis and interactive dashboards.

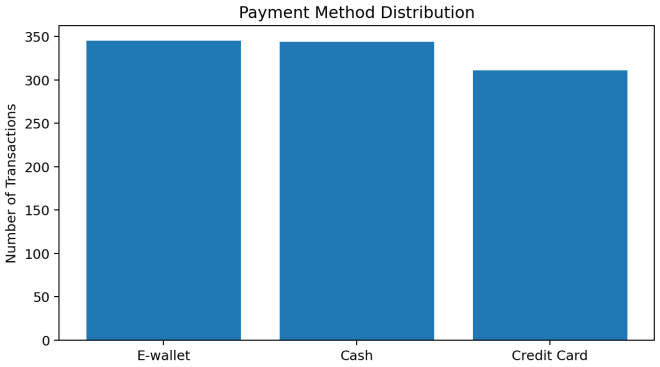
OUTPUT:

PYTHON:

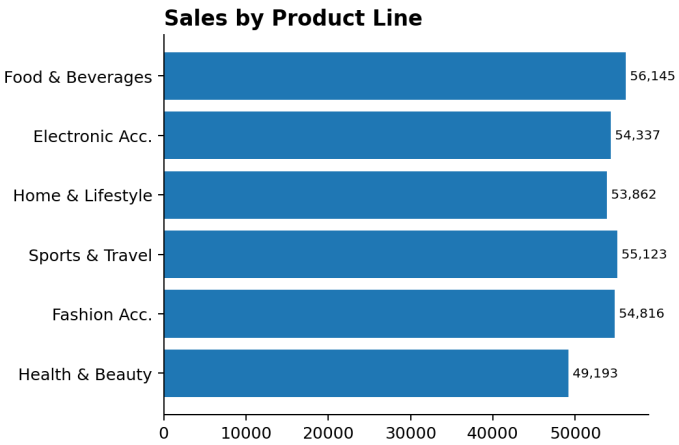


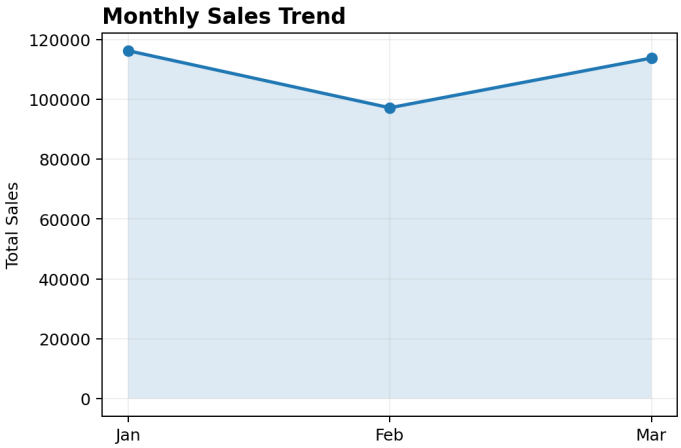


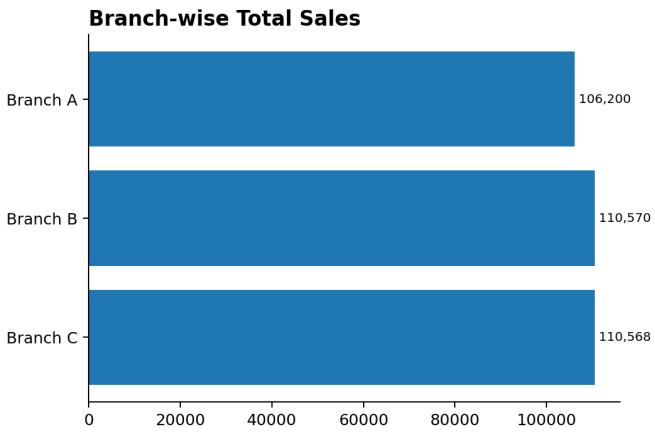

Monthly Sales Trend
Total Sales
115000
112500
110000
107500
105000
102500
100000
97500
Jan
Feb
Mar
Month

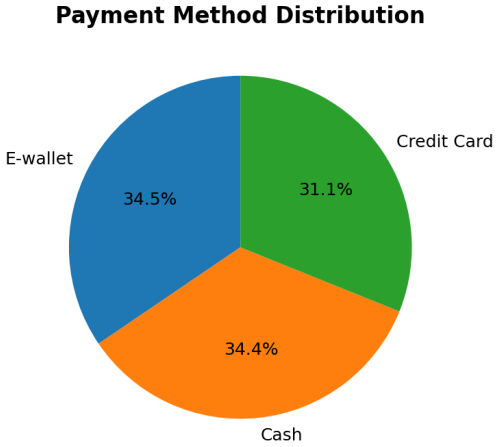

Payment Method Distribution
Number of Transactions
350
300
250
200
150
100
50
0
E-wallet
Cash
Credit Card

POWER-BI

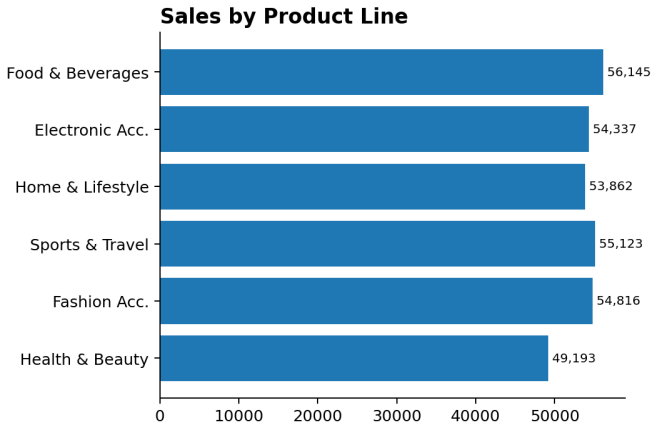

Sales by Product Line
Food & Beverages
Electronic Acc.
Home & Lifestyle
Sports & Travel
Fashion Acc.
Health & Beauty
56,145
54,337
53,862
55,123
54,816
49,193
0
10000
20000
30000
40000
50000

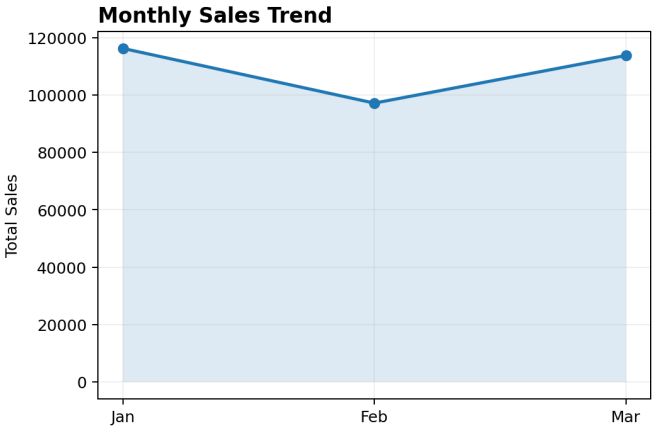

Monthly Sales Trend
Total Sales
120000
100000
80000
60000
40000
20000
0
Jan
Feb
Mar

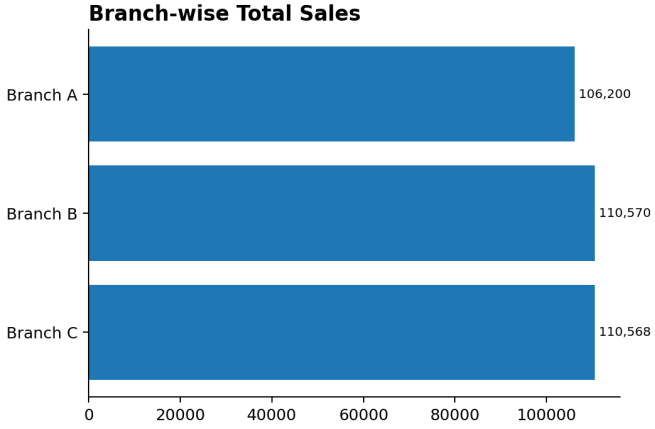

Branch-wise Total Sales
Branch A
Branch B
Branch C
106,200
110,570
110,568
0
20000
40000
60000
80000
100000

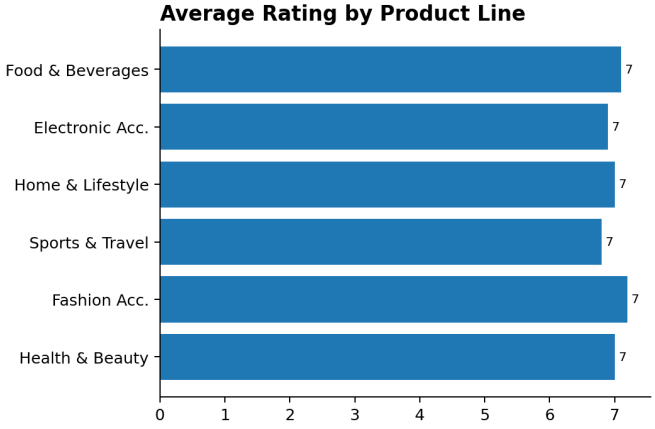

Payment Method Distribution
E-wallet
Credit Card
Cash
34.5%
31.1%
34.4%

TABLEAU


Sales by Product Line
Food & Beverages
Electronic Acc.
Home & Lifestyle
Sports & Travel
Fashion Acc.
Health & Beauty
56,145
54,337
53,862
55,123
54,816
49,193
0
10000
20000
30000
40000
50000


Monthly Sales Trend
Total Sales
120000
100000
80000
60000
40000
20000
0
Jan
Feb
Mar


Branch-wise Total Sales
Branch A
Branch B
Branch C
106,200
110,570
110,568
0
20000
40000
60000
80000
100000


Average Rating by Product Line
Food & Beverages
Electronic Acc.
Home & Lifestyle
Sports & Travel
Fashion Acc.
Health & Beauty
7
7
7
7
7
7
0
1
2
3
4
5
6
7